

DISSERTATION CASE STUDY
ON
**SRS FOR E-TRADE MANAGEMENT
SYSTEM ARCHITECTURE**

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1. Introduction

1.1 Purpose

The purpose of the system is to support management of electronic trading by providing functionality for customer interaction, order processing, delivery management, and enterprise business process realization. This is explicitly stated as the objective of the proposed e-trade management system architecture.

1.2 Scope

The system supports electronic trading through two functional areas:

- Landing Web-pages Management Unit (customer interaction and order acquisition)
- Web-enterprise Management Unit (order processing and enterprise management)

The paper describes the functional responsibilities of these areas but does not define organizational scope beyond electronic trading.

1.3 Intended Users

Explicitly mentioned users include:

- Consumer
- Buyer
- Customer
- Online store managers (customer service managers)
- Suppliers
- Manufacturers

No additional user roles are specified.

1.4 System Overview

The system supports:

- presentation of products and services
- customer interaction
- order creation
- order processing
- financial processing
- resource management
- supply management
- delivery management

- enterprise business process support

The paper presents these as functional responsibilities of the e-trade management system.

2. Functional Requirements

ID	Description
FR-01	The system shall interact with users through landing web pages.
FR-02	The system shall receive customer orders.
FR-03	The system shall collect primary information associated with customer orders.
FR-04	The system shall process customer orders.
FR-05	The system shall support delivery of customer orders.
FR-06	The system shall support enterprise business process realization.
FR-07	The system shall generate shop window data using product assortment information and business rules.
FR-08	The system shall provide information about goods and services to users.
FR-09	The system shall provide detailed descriptions of goods and services.
FR-10	The system shall provide graphical representations of products.
FR-11	The system shall gather information about goods and services.
FR-12	The system shall organize and structure information about goods and services.
FR-13	The system shall update information about goods and services.
FR-14	The system shall process information about goods and services.
FR-15	The system shall harmonize data representation formats.
FR-16	The system shall track changes to product and service information.
FR-17	The system shall provide complete information on product and service characteristics.
FR-18	The system shall support demand formation for goods and services.
FR-19	The system shall prepare advertising and marketing information.
FR-20	The system shall monitor consumer reviews.
FR-21	The system shall monitor statistics related to advertising and marketing.
FR-22	The system shall determine advertising and marketing effectiveness.

FR-23	The system shall inform consumers.
FR-24	The system shall support personal interaction with customers.
FR-25	The system shall personalize information according to customer interests.
FR-26	The system shall support customer loyalty management.
FR-27	The system shall allow consumers to select goods.
FR-28	The system shall allow consumers to review purchase information and purchase terms.
FR-29	The system shall identify consumers during ordering.
FR-30	The system shall capture delivery address information.
FR-31	The system shall allow consumers to choose payment terms.
FR-32	The system shall allow consumers to choose delivery terms.
FR-33	The system shall allow consumers to confirm their selections.
FR-34	The system shall create buyer orders.
FR-35	The system shall provide order-related information.
FR-36	The system shall monitor markets for goods and services.
FR-37	The system shall monitor supplier warehouse load.
FR-38	The system shall monitor prices for goods and services.
FR-39	The system shall support price formation decisions.
FR-40	The system shall support assortment formation decisions.
FR-41	The system shall update assortment data.
FR-42	The system shall verify incoming order information.
FR-43	The system shall track execution status of customer orders.
FR-44	The system shall determine order execution conditions.
FR-45	The system shall determine conditions for working with goods.
FR-46	The system shall calculate order execution time.
FR-47	The system shall support financial interaction required for trading operations.
FR-48	The system shall monitor financial resources on accounts.
FR-49	The system shall allocate financial resources on accounts.
FR-50	The system shall eliminate shortages of goods and services.
FR-51	The system shall account for resources.
FR-52	The system shall collect current order processing information.
FR-53	The system shall analyze current order processing information.
FR-54	The system shall collect statistical relationship data with counterparties.
FR-55	The system shall support management decision making using operational and statistical information.
FR-56	The system shall support accounting for stock used in order fulfillment.

FR-57	The system shall manage customer order delivery.
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Derived Requirements: None.

3. Non-Functional Requirements

ID	Category	Description
NFR-01	Scalability	The system shall support scalability.
NFR-02	Maintainability	The system shall support enhancement and modernization.
NFR-03	Interoperability	The system shall support platform-independent service interaction.
NFR-04	Availability	The system shall make operational data available for online operation.
NFR-05	Performance	The system shall support real-time event handling.
NFR-06	Maintainability	The system shall allow modification of the information system set without modifying existing services.

No explicit requirements were identified for:

- Reliability
- Security
- Portability
- Usability

Therefore:

Not Specified

4. Actors

Explicitly mentioned actors:

- Consumer
 - Buyer
 - Customer
 - Supplier
 - Manufacturer
 - Customer service manager
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5. External Systems

Explicitly mentioned:

- Accounting system
- Bank
- Payment system

No additional external systems are explicitly required.

6. Business Rules

BR-01. An order consists of the entities **user**, **order**, and **terms of service**. Evidence: Section II (Ordering process).

BR-02. Orders are associated with the specific online store and shop window where they are created. Evidence: Section II.

No additional explicit business rules are stated.

7. Assumptions

Not Specified.

8. Constraints

Explicit constraints:

- Business requirements determine operating conditions of the web store, including assortment formation, payment terms, delivery terms, trading conditions, marketing activities, and showcase content. Evidence: Section II – Web-enterprise management.

No additional constraints are explicitly specified.

9. Out of Scope

Not Specified.